Risk-controlled post-hoc rescue of rare cell types in single-cell RNA-seq under label scarcity

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Abstract

Motivation: Rare cell types in single-cell RNA-seq are systematically missed by semi-supervised classifiers when only a handful of rare cells are labelled. Simply encouraging more rare-class predictions recovers some of these cells but inflates false positives, which can be more damaging than the original omissions.

Results: We present scRareRefine, a post-hoc module that refines the predictions of a frozen scANVI backbone. It scores each cell by an anisotropic prototype-membership distance computed only on the training set, and decides which cells to relabel through a conformal threshold and three abstention gates calibrated only on a validation set, so that the false-rescue rate (FFR) is held below a fixed budget $\alpha = 0.01$. Across six datasets, nine methods and three random splits, scRareRefine raises mean rare-cell F1 in the label-scarce regime from 0.72 (scANVI) to 0.88 while keeping the worst-case FFR at $0.0098 \leq \alpha$; it is the only method that achieves both, and it never underperforms scANVI in this regime (paired one-sided Wilcoxon $p = 1.3 \times 10^{-6}$). The method is conditional by design: it abstains, returning the backbone prediction, whenever rescue is not warranted, and we characterise its failure modes, including a recall ceiling for transcriptionally entangled rare types.

Availability: Code and reproduction scripts are available at `https://github.com/REPO`.

1 Introduction

Single-cell RNA sequencing (scRNA-seq) routinely resolves the major cell types of a tissue, but the populations that matter most for a biological question are often the *rare* ones: a small dendritic-cell subset, a sparse endocrine population, or a handful of mast cells. Automated annotation methods trained with only a few labelled examples of such a type tend to absorb its cells into a transcriptionally adjacent majority class. Semi-supervised models such as scANVI [1], which combine a variational backbone with a small amount of label information, are a strong default for this task, yet they inherit the same difficulty: when the labelled budget for a rare type is small, the type is under-represented in the supervised signal and is systematically missed at test time.

A natural reflex is to make the classifier more willing to predict the rare class. This does recover some missed cells, but it also relabels majority cells that merely sit near the rare population, inflating false positives. In many downstream analyses a false rare call (e.g. reporting mast cells where there are none) is more costly than a missed one. The practical problem is therefore not simply "predict more rare cells" but "recover missed rare cells *without* paying an uncontrolled false-positive cost" — and to do so *inductively*, using only information available at training and calibration time.

We address this with **scRareRefine**, a lightweight post-hoc module that refines, rather than replaces, the predictions of a frozen scANVI backbone (Fig. 1). scRareRefine first builds a class prototype for every labelled training type in the scANVI latent space and scores each query cell by an anisotropic membership distance to the rare prototype, normalised by each class's compactness. It then decides which cells to relabel through a calibration step performed entirely on a held-out validation set: a conformal threshold on validation non-rare scores controls the false-rescue rate, and three abstention gates — a separability safety net, a necessity guard, and a validation-adaptive candidate-rank selector — prevent the module from acting when rescue is unsafe or unnecessary. The reference prototypes come only from the training set and every threshold comes only from the validation set; test labels are used exclusively for final evaluation.

The central design choice is to trade a little recall for a *controlled* false-rescue rate (FFR). We fix the FFR budget to a publication-grade $\alpha = 0.01$ across all datasets and never tune it, so that improvements in rare-cell F1 are never bought with an uncontrolled increase in false positives. Because the module abstains when its gates are not satisfied, it reduces exactly to the backbone on datasets where rescue is not warranted; its benefit is concentrated in the genuinely label-scarce, genuinely-missed regime.

Contributions.

1. **A risk-controlled rescue module.** scRareRefine is a fully inductive, post-hoc refinement of a frozen scANVI classifier that recovers missed rare cells while holding the false-rescue rate below a fixed budget $\alpha = 0.01$ (Section 2).

2. **Label-scarce gains with controlled FFR.** Across six datasets, nine methods and three splits, scRareRefine attains the highest mean rare-cell F1 in the label-scarce regime (0.88 vs. 0.72 for scANVI) and is the only method whose worst-case FFR stays $\leq \alpha$; it never underperforms scANVI in this regime (paired $p = 1.3 \times 10^{-6}$) (Section 3.2).

3. **Component analysis.** A leave-one-out ablation shows that the adaptive candidate-rank selector is the component that drives F1, the conformal threshold is what controls FFR, and the two gates act as safety mechanisms that prevent regressions rather than as performance boosters (Section 3.3).

4. **An honest analysis of the separability gate.** Through a controlled stress test and a benchmark-wide sensitivity sweep we show that the gate threshold is a deliberately conservative, pre-specified prior whose risk axis is dataset-dependent, and we report where it is too cautious (Section 3.4).

5. **Characterised failure modes.** We delineate the regime in which the method is a no-op and its concrete limitations, including a recall ceiling for rare types that are geometrically entangled with a majority population (Section 4).

The remainder of the paper introduces related work, details the method (Section 2), presents the experiments (Section 3), and discusses limitations and outlook (Section 4).

Related work. *Reference-based annotation* methods — logistic-regression CellTypist [2], imbalance-aware scBalance, and the interpretable transformer TOSICA [3] — transfer labels from a reference but do not provide an explicit false-positive guarantee for a target rare type, and their accuracy degrades when the rare type is sparsely labelled. *Rare-population discovery* methods such as scCAD detect anomalous clusters in an unsupervised manner; mapping such discoveries onto a *specific* rare type incurs an inherent precision penalty. *Prototype and semi-supervised* approaches, including the scVI/scANVI family [1, 4] and prototype classifiers, motivate our latent-space prototype scoring but do not target the scarce-label rescue problem with risk control. Finally, *conformal prediction* [5, 6] provides the distribution-free machinery we use to translate a validation calibration set into a finite-sample bound on the false-rescue rate. scRareRefine combines latent-space prototype geometry with conformal calibration to obtain a rescue that is both effective in the scarce regime and risk-controlled, a combination not offered by prior methods.

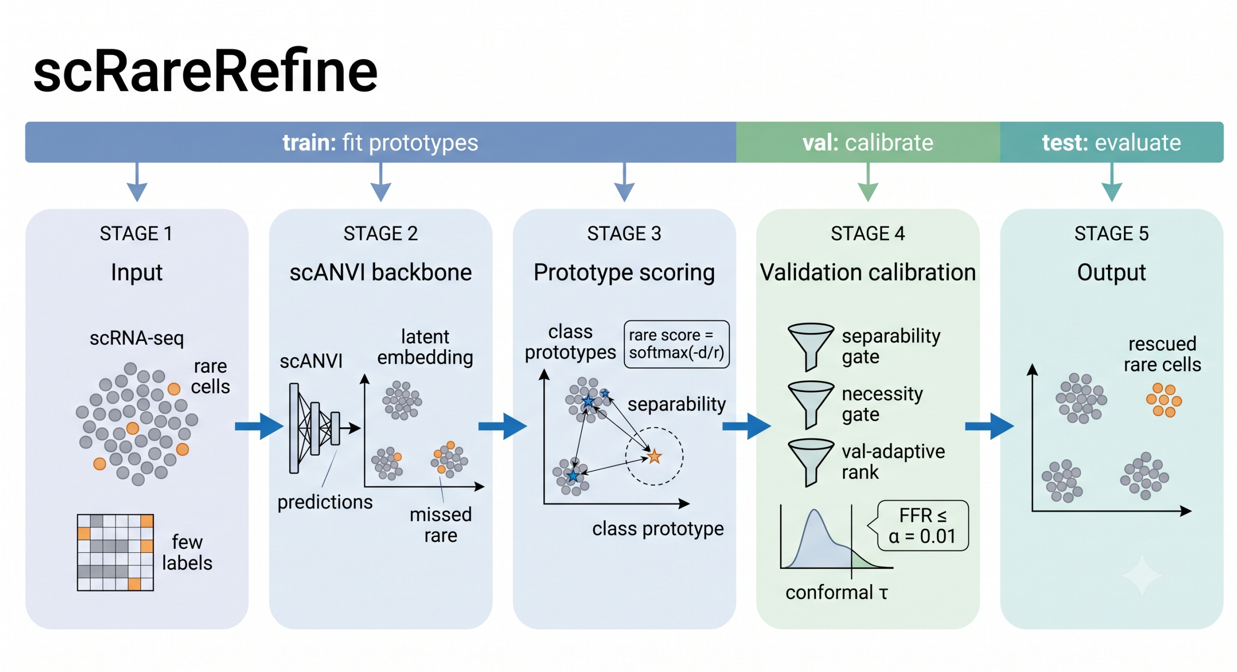

Figure 1: **Overview of scRareRefine.** A frozen scANVI backbone produces a latent embedding and a prediction for every cell. On the training set we build class prototypes; for each query cell we compute an anisotropic rare-membership score. A validation set calibrates a conformal threshold and three abstention gates so that the false-rescue rate stays below α. At test time, candidate cells whose score exceeds the threshold are relabelled to the rare class. Reference prototypes use only training cells; all thresholds use only validation cells.

2 Methods

2.1 Setup and inductive protocol

Let a dataset contain cells with expression profiles and a single designated rare type r. We split cells into training, validation and test partitions using a batch-/donor-held-out scheme (target $70/15/15$), so that validation and test cells come from donors unseen during training; this is the harder, more realistic inductive setting. A semi-supervised scANVI model [1], itself initialised from an scVI model [4], is trained on the training partition with all majority cells labelled and only a limited budget of rare cells labelled. The labelled rare budget is $\max(5, \lfloor p\, N^r_{\text{tr}} \rfloor)$ for a nominal fraction p, where N^r_{tr} is the number of rare cells in the training partition; the floor of five reflects the minimum needed to estimate a prototype. The model yields, for every cell, a latent embedding $z \in \mathbb{R}^d$ and a predicted label. scRareRefine consumes these frozen outputs and never retrains the backbone.

2.2 Prototype-membership scoring

For each class c observed among the *labelled* training cells we form a prototype $\mu_c = \frac{1}{|S_c|} \sum_{i \in S_c} z_i$, the mean latent of its labelled cells S_c, and a robust radius $r_c = \max\big(\text{median}_{i \in S_c} \|z_i - \mu_c\|, \epsilon\big)$ (set to a conservative default when $|S_c| < 3$). For a query cell with embedding z we define an *anisotropic rare-membership score*

$$s(z) = \text{softmax}_c\Big(- \frac{\|z - \mu_c\|}{r_c} \Big)\big[r\big], \tag{1}$$

i.e. the rare-class component of a softmax over per-class distances normalised by each class's compactness. The per-class normalisation lets the score adapt to classes of different spread and limits intrusion from a tight neighbouring majority cluster. We also define a train-only *separability ratio*

$$\rho \;=\; \frac{\min_{c\neq r}\lVert \mu_r - \mu_c \rVert}{\frac{1}{|S_r|}\sum_{i\in S_r}\lVert z_i - \mu_r \rVert}, \tag{2}$$

the distance from the rare prototype to its nearest majority prototype divided by the mean rare intra-class radius; ρ measures how geometrically distinct the rare type is, using only training information.

2.3 Conformal calibration and abstention gates

Given the frozen scores, scRareRefine decides which test cells to relabel using only validation labels. Three gates and one threshold are applied in sequence (Algorithm 1).

Separability gate. If $\rho < \rho_{\text{low}}$ the rare and majority populations are too entangled for prototype rescue to be safe and the module abstains. We fix $\rho_{\text{low}} = 1.3$ as a dataset-independent prior (Section 3.4).

Necessity guard. If the validation backbone already recovers the rare type — specifically if the number of validation rare cells it misses is below M — rescue is unnecessary and may only add noise, so the module abstains. We fix $M = 3$, the minimum support needed for the validation-adaptive selection below to carry signal.

Conformal threshold. Let the validation non-rare scores be $s_1, \ldots, s_n$. We set the threshold τ to the finite-sample $(1-\alpha)$ conservative order statistic $\tau = s_{(k^\star)}$ with $k^\star = \lceil (1-\alpha)(n+1) \rceil$ (and $\tau = +\infty$, i.e. no rescue, if $k^\star > n$). Under exchangeability of validation and test non-rare cells, relabelling only cells with $s \geq \tau$ bounds the probability that a non-rare cell is rescued at α.

Validation-adaptive candidate rank. A cell is a *candidate* only if the rare prototype is among its k nearest prototypes (isotropic Euclidean rank $\leq k$) and its backbone label is not already rare. We choose $k \in \{1, 2, 3\}$ on validation as the value that maximises validation rare-cell F1 subject to a Wilson 95% upper bound on the validation FFR not exceeding α, breaking ties toward the smaller (more conservative) k. Using the Wilson upper bound rather than a point estimate guards against the validation/test distribution shift induced by donor-held-out splits.

Rescue rule. A test cell is relabelled to r iff it is a candidate (rank $\leq k$) *and* $s \geq \tau$. All other predictions are left unchanged.

2.4 False-rescue rate

We quantify the cost of rescue by the *false-rescue rate* $\text{FFR} = \big|\{\text{non-rare test cells relabelled to } r\}\big| \,/\, \big|\{\text{non-rare test cells}\}\big|$. The conformal threshold and gates are designed so that FFR is controlled at the fixed budget $\alpha = 0.01$. Because validation and test cells come from different donors under our split, this control is empirical rather than a strict finite-sample guarantee; we report FFR with raw counts and treat α as a fixed prior, never tuned per dataset.

Algorithm 1 scRareRefine conformal rescue (inductive)

Require: frozen latents and predictions for train/val/test; rare type r; budget α

fit prototypes $\{\mu_c\}$, radii $\{r_c\}$, separability ρ on labelled *train* cells

if $\rho < \rho_{\text{low}}$ **then return** backbone predictions ▷ separability gate

end if

$v_{\text{miss}} \leftarrow$ #validation rare cells missed by backbone

if $v_{\text{miss}} < M$ **then return** backbone predictions ▷ necessity guard

end if

compute scores Eq. (1) for val and test; set τ from val non-rare scores

choose $k \in \{1, 2, 3\}$ maximising val rare-F1 s.t. Wilson-upper val-FFR $\leq \alpha$

for each test cell: relabel to r if rank$\leq k$ and $s \geq \tau$

return refined test predictions

Table 1: **Per-dataset rare-cell F1 in the label-scarce regime** (mean over fractions $p \leq 0.10$ and three seeds). scRareRefine is the best or tied-best inductive method on every dataset while keeping its worst-case FFR within the budget $\alpha = 0.01$. "Best baseline" excludes the transductive HiCat reference.

Dataset (rare type)	scANVI	Best baseline	**scRareRefine**	$\text{FFR}_{\max}$
immune dendritic (ASDC)	0.59	0.72 (kNN)	**0.94**	0.0010
Baron pancreas (gamma)	0.53	0.46 (kNN)	**0.66**	0.0098
lung (lymphatic EC)	0.72	0.92 (kNN)	**0.98**	0.0029
small intestine (tuft)	0.98	0.98 (kNN)	**0.98**	0.0005
stomach (mast)	0.61	0.49 (CellTypist)	**0.72**	0.0002
integrated pancreas (endo.)	0.89	0.98 (kNN)	**1.00**	0.0006

3 Results

3.1 Experimental setup

We evaluate on six scRNA-seq datasets, each with a designated rare type: human immune dendritic cells (ASDC), Baron pancreas (gamma), an integrated human pancreas (endothelial), and three Tabula Sapiens subsets — lung (lymphatic endothelial), small intestine (tuft) and stomach (mast). For every dataset we use a donor-held-out $70/15/15$ split and vary the labelled rare fraction over $\{0.01, 0.05, 0.10, \text{all}\}$, repeating each configuration over three random seeds $(42, 43, 44)$. We compare scRareRefine against eight methods that read the same split and the same labelled subset: the scANVI backbone, a k-NN classifier on the scANVI latent, CellTypist, scBalance, ProtoCloud, scCAD, TOSICA, and HiCat. HiCat is transductive (its dimensionality reduction is fit on combined train and test cells) and is therefore reported as an upper-bound reference rather than a fair inductive competitor; TOSICA is run with its default pathway configuration. We report rare-cell F1 and the false-rescue rate (FFR), aggregated as mean $\pm$ standard deviation over seeds. Because the labelled budget is $\max(5, \lfloor p\, N^r_{\text{tr}} \rfloor)$, several nominal fractions collapse to the same labelled set on small-rare datasets; we therefore report the label-scarce comparison over the 15 *distinct* labelled configurations rather than the 18 nominal ones.

3.2 scRareRefine improves rare-cell F1 in the scarce regime with controlled FFR

Across the label-scarce regime scRareRefine attains the highest mean rare-cell F1 of all nine methods (Fig. 2a): 0.88, against 0.72 for the scANVI backbone and 0.73 for the next-best inductive method (k-NN on the scANVI latent); the remaining methods range from 0.48 to 0.65, and the transductive HiCat

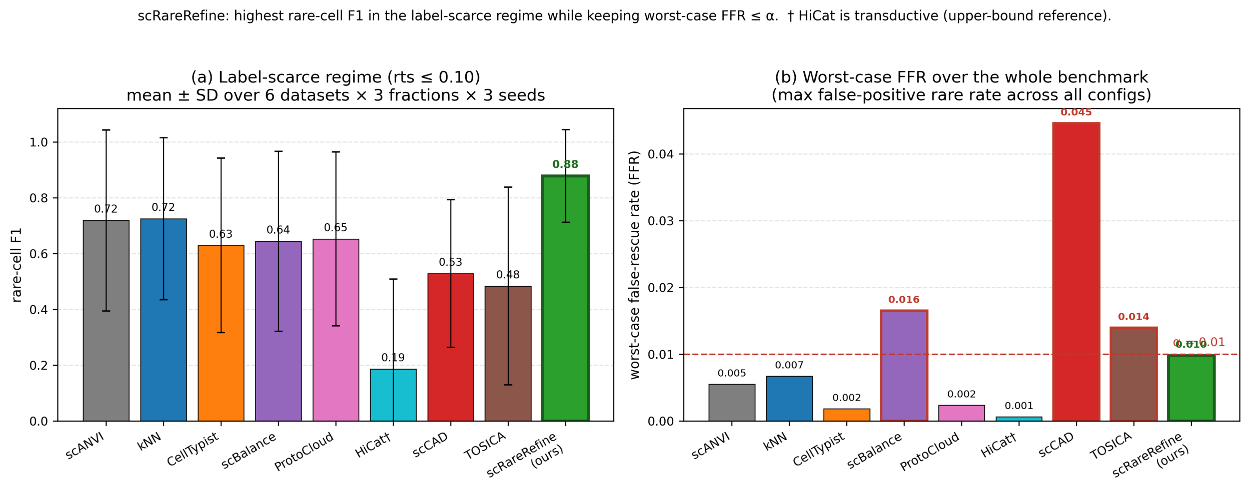

Figure 2: **Main result.** (a) Mean rare-cell F1 in the label-scarce regime ($p \leq 0.10$), averaged over six datasets, three fractions and three seeds. (b) Worst-case FFR over the whole benchmark; the dashed line is the budget $\alpha = 0.01$. scRareRefine is the only method that combines the highest scarce-regime F1 with a worst-case FFR $\leq \alpha$. † HiCat is transductive.

reference is lower (0.19) because it returns no rare cells on several datasets. Per dataset (Table 1), scRareRefine is the best or tied-best inductive method on all six datasets, with the largest gains where the backbone is weakest (immune dendritic cells, $0.59 \to 0.94$; stomach mast cells, $0.61 \to 0.72$).

This F1 advantage is not bought with an uncontrolled false-positive cost. scRareRefine is the only method whose worst-case FFR over the whole benchmark stays within budget ($0.0098 \leq \alpha$, Fig. 2b); by contrast scCAD (0.045), scBalance (0.016) and TOSICA (0.036) exceed it. A paired analysis over the 54 scarce configurations (six datasets $\times$ three distinct fractions $\times$ three seeds) shows that scRareRefine never underperforms the backbone: 29 wins, 25 ties — the ties are abstentions — and 0 losses, with a mean F1 improvement of $+0.16$ (bootstrap 95% CI $[+0.09, +0.24]$). Against every one of the eight baselines the improvement's confidence interval excludes zero (one-sided Wilcoxon $p \leq 1.3 \times 10^{-6}$), which we read as directional evidence given the correlated paired cells. Because the labelled budget has a floor of five cells, the 18 nominal scarce configurations collapse to 15 distinct ones; over these, scRareRefine beats the majority of baselines in $15/15$ and is the single best method in $14/15$ (Supplementary S3).

3.3 Component ablation

The full method raises mean rare-cell F1 from 0.76 (backbone) to 0.89 (Fig. 3, left). Removing each component in turn separates two roles. The validation-adaptive candidate-rank selector is the only component that materially raises F1: replacing it with the most conservative fixed rank ($k = 1$) lowers F1 by 0.010. The conformal threshold and the two gates, by contrast, barely change mean F1 but are what keep the method safe. Removing the conformal threshold leaves mean F1 essentially unchanged ($+0.002$) yet pushes the worst-case FFR to 0.0165, above budget; removing the separability gate even raises mean F1 (by 0.018) but likewise breaks the budget (worst-case FFR 0.0153); and removing the necessity guard leaves the benchmark mean unchanged but causes per-dataset regressions on already-saturated datasets (integrated pancreas, $0.990 \to 0.977$, with a non-zero FFR appearing). The rank-sensitivity panel (Fig. 3, right) confirms the value of adapting k: every fixed choice is dominated in F1 by the adaptive selector (k=1: 0.877, k=2: 0.865, k=3: 0.853 vs. adaptive 0.887), and the most permissive fixed rank (k=3) drives the worst-case FFR to 0.046, far above budget, whereas the adaptive selector stays within it (0.0098). In summary, the adaptive

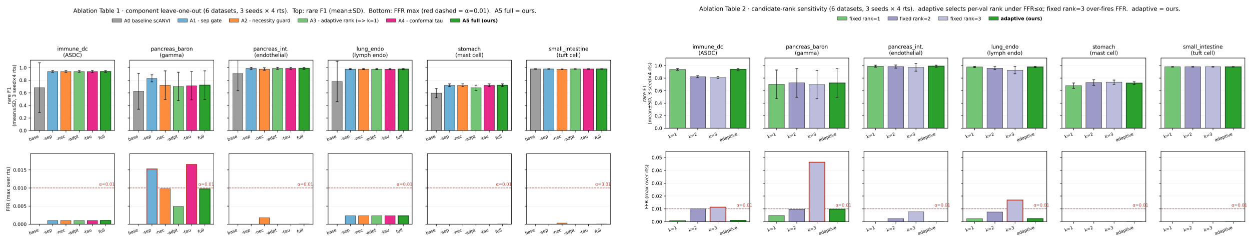

Figure 3: **Ablation.** Left: leave-one-out over the four components (F1 top, worst-case FFR bottom). Right: candidate-rank sensitivity (fixed k vs. validation-adaptive). Removing the conformal threshold or the separability gate barely changes mean F1 but breaks the FFR budget; the adaptive rank is the component that raises F1; a fixed rank $= 3$ over-fires FFR.

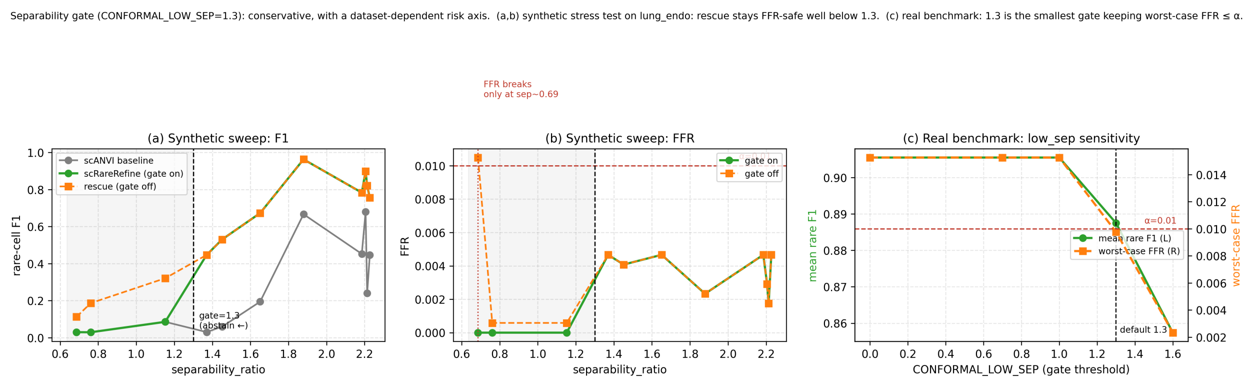

Figure 4: **Separability-gate analysis.** (a,b) Controlled semi-synthetic stress test (entangling a rare type toward its nearest majority): rescue stays FFR-safe well below the gate at 1.3. (c) Benchmark-wide sensitivity of the gate threshold: 1.3 is the smallest threshold that keeps worst-case FFR $\leq \alpha$ on real data.

rank drives F1, the conformal threshold controls FFR, and the two gates prevent regressions.

3.4 The separability gate is a conservative, dataset-dependent prior

The separability gate uses a fixed, dataset-independent threshold ($\rho_{\text{low}} = 1.3$). We probe this choice from two directions (Fig. 4). First, a controlled semi-synthetic stress test progressively entangles a rare population (lung lymphatic endothelial cells) toward its nearest majority type; rescue *without* the gate stays within the FFR budget down to separability ≈ 0.76 and produces a single marginal violation (18 false rescues out of 1716, FFR 0.0105) only at the lowest separability (≈ 0.69), well below the gate (Fig. 4a,b). Second, a benchmark-wide sweep of the gate threshold (Fig. 4c) shows that 1.3 is the smallest threshold that keeps the worst-case FFR within budget on real data: lowering it to ≤ 1.0 raises mean F1 by 0.018 but admits two FFR violations (worst-case 0.0153, on the low-separability Baron-pancreas configurations at $\rho \approx 1.22$), whereas raising it to 1.6 only sacrifices F1 ($0.888 \rightarrow 0.857$) with no further FFR benefit. The two experiments disagree on where rescue becomes unsafe (≈ 1.22 on real Baron pancreas vs. ≈ 0.69 on the synthetic sweep), which is the central observation: separability alone does not monotonically determine rescue risk. We therefore present 1.3 not as a located collapse boundary but as a deliberately conservative, pre-specified prior chosen by the FFR constraint rather than tuned for F1 — lowering it would in fact *raise* mean F1 — that occasionally abstains where rescue would have been safe.

3.5 Qualitative rescue in latent space

Fig. 5 visualises the rescue on immune dendritic cells in the scANVI latent space. Of the true rare cells (Fig. 5a), the backbone recovers those forming a compact cluster (recall 0.72, panel b). scRareRefine relabels 23 further cells lying at the edge of this cluster, raising recall to 0.90 at precision 0.99, with a single false rescue (panels c,d). The 13 rare cells that remain missed are scattered far from the rare prototype; admitting them would require widening the candidate pool past the FFR budget. This is the geometric recall ceiling discussed in Section 4.

4 Discussion

scRareRefine is a conservative, risk-controlled rescuer, and its value is concentrated in a specific regime rather than uniform across all settings. We make this scope explicit.

The method is conditional by design. Two abstention gates cause scRareRefine to return the unmodified backbone prediction whenever the rare and majority prototypes are too entangled or the validation backbone already recovers the rare type. On datasets where the backbone is already saturated, the module therefore reduces exactly to scANVI — it neither helps nor harms — and its benefit is restricted to the genuinely label-scarce, genuinely missed regime. This is a feature for safety, but it means the method should not be read as a universal improvement.

A recall ceiling for entangled rare types. For a rare type that is transcriptionally interleaved with a majority population, prototype-distance rescue has an intrinsic ceiling: a substantial fraction of true rare cells fall at candidate rank ≥ 3, where admitting them would require widening the candidate pool past the point at which the FFR budget is exceeded. The adaptive-rank selector correctly refuses to do so, so recall plateaus (Fig. 5). Breaking this ceiling likely requires expression-side signals (e.g. marker genes) and is left to future work.

FFR control is empirical and can sit near the boundary. The conformal threshold provides empirical FFR control under validation calibration, not a strict finite-sample guarantee on test: under donor-held-out splits the exchangeability assumption is only approximately satisfied, and on one dataset the test FFR reaches the $\alpha = 0.01$ budget. We mitigate this with a Wilson upper bound for rank selection and report all FFR values with raw counts, but the budget should be read as empirical control under distribution shift.

Seed sensitivity at extreme scarcity. At the smallest budget (five labelled rare cells) the improvement on one dataset is positive on average but unstable across seeds, mirroring the variance of the backbone itself; the instability disappears once at least ten rare cells are labelled. We therefore do not claim stable improvement at the five-label point on that dataset.

Evaluation caveats. Because the labelled budget has a floor of five, several nominal label fractions collapse to the same labelled set on small-rare datasets; we report distinct configurations to avoid double-counting. Our significance tests pair predictions by (dataset, fraction, seed); these cells are correlated across seeds and near-duplicated across collapsed fractions, so the p-values are optimistic and are best read as a robust paired improvement with an effect size whose confidence interval excludes zero. We evaluate six predominantly human single-cell datasets and do not test cross-species or disease-state transfer, both of which would further stress the exchangeability assumption. Finally, scRareRefine operates on scANVI embeddings; whether the same conformal rescue adds value on top of single-cell foundation-model embeddings is untested.

Outlook. The combination of latent-space prototype geometry with conformal calibration offers a general template for risk-controlled post-hoc refinement. Natural extensions include expression-side scoring to address the recall ceiling, batch-conditional calibration to tighten FFR control under donor shift, and application on top of foundation-model embeddings.

5 Conclusion

We introduced scRareRefine, a fully inductive post-hoc module that refines a frozen scANVI classifier to recover rare cell types under label scarcity while keeping the false-rescue rate below a fixed budget. By scoring cells with training-only prototype distances and gating relabelling with validation-only conformal calibration, the method raises rare-cell F1 in the label-scarce regime — the only method in our benchmark to do so with a controlled worst-case false-rescue rate — and abstains when rescue is not warranted. We further provided an honest analysis of its conservative separability gate and its failure modes. Risk-controlled refinement is a practical way to recover rare populations without paying an uncontrolled false-positive cost, and we hope it encourages reporting the operating point, not only the accuracy, of rare-cell annotation methods.

References

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A Supplementary material

[DRAFT.] *The supplementary section will collect the full benchmark grid, the paired significance table, the distinct-configuration accounting, additional separability and robustness analyses, runtime, additional qualitative panels, and reproduction provenance.*

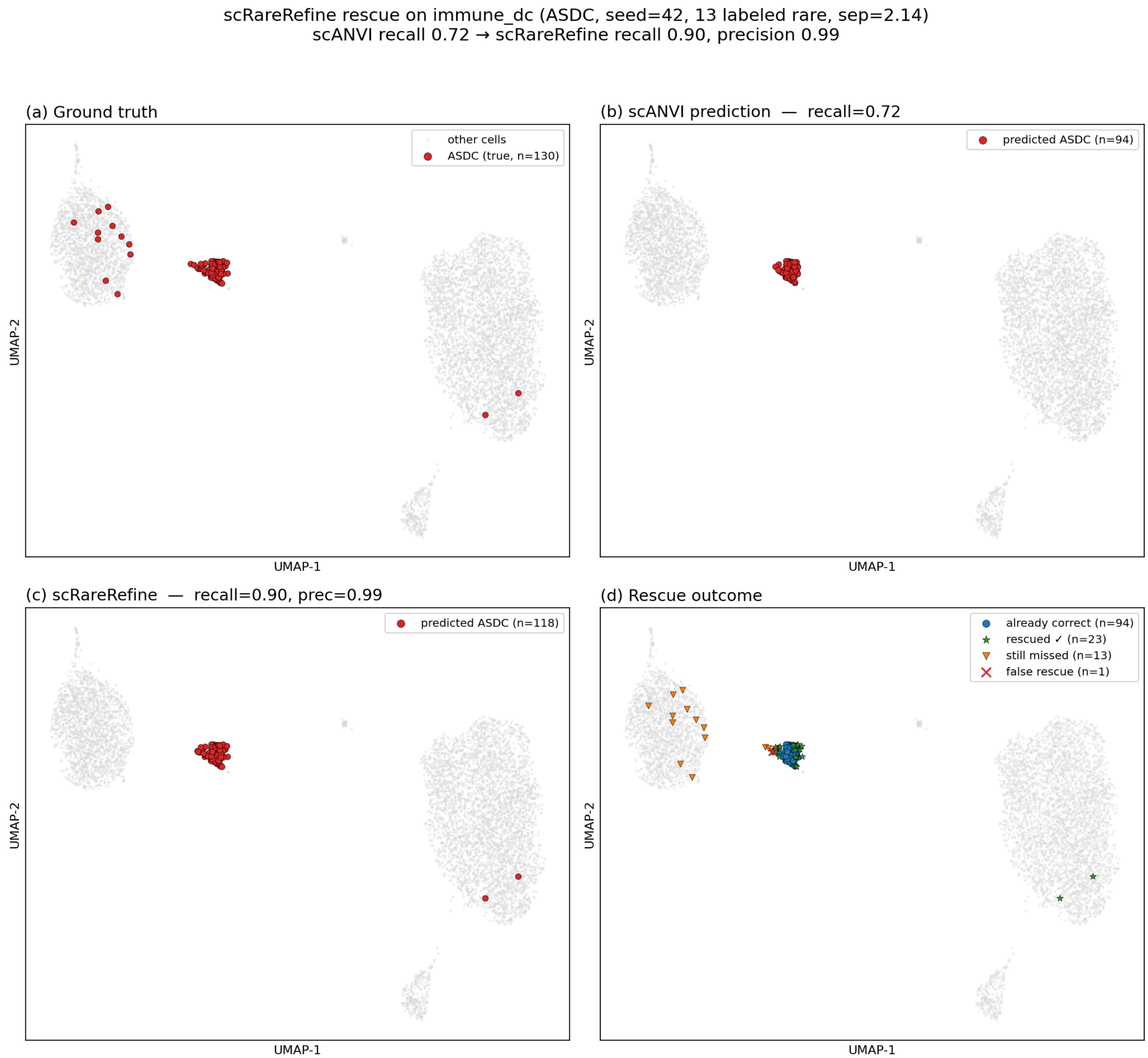

Figure 5: **Rescue in the scANVI latent space (immune dendritic cells).** Cells missed by scANVI that lie near the rare cluster are recovered (recall $0.72 \rightarrow 0.90$), whereas geometrically distant rare cells remain missed, illustrating the recall ceiling discussed in Section 4.